

Santrupt Potphode

+91-9823541475 | santrupt.potphode29@gmail.com | linkedin.com/in/santrupt29 | github.com/santrupt29 | santrupt.dev

Professional Summary

Backend Developer specializing in high-performance **distributed systems** using **Java**, **Spring Boot**, and **Microservices**. Experienced in **LangGraph AI** agents, **scalable cloud workflows** and **real-time applications**.

Education

Veermata Jijabai Technological Institute

B.Tech in Electronics and Telecommunication Engineering

CGPA: 7.70/10.0

Aug 2023 - May 2027

Mumbai, Maharashtra

Experience

Software Development Engineer Intern

May 2026 - July 2026

Swastik Software Technologies

Mumbai, Maharashtra

- Orchestrated a **LangGraph AI** agent handling **11 care-ops workflows**, **multi-turn state**, and **3-role RBAC** for a production NDIS platform.
- Engineered **3-tier LLM routing** with **5-stage timing instrumentation**, eliminating **60s+** latency that caused production 502 errors.
- Extended **Spring Boot** for recurring shift series with **25-occurrence @Transactional batches**; wired Python NLP agent.
- Designed a **5-table MySQL** observability schema across Java and Python services with **aiomysql** pooling and **Metabase** dashboard.

Projects

Doodle-Sync - Real-Time Multiplayer Drawing Game 🎮 *Spring Boot 3.5, Kafka, Redis, WebSocket, Docker*

- Architected **7 microservices** via **Spring Cloud Gateway**, using **Resilience4j** circuit breakers for fault-tolerant **Feign** calls.
- Achieved **p99 < 1ms latency** via **WebSocket STOMP**, routing strokes through **Redis Pub/Sub** while reserving **Kafka** for durable events.
- Implemented a 5-state game machine with **time-decay scoring** and **Levenshtein-based** near-miss guess detection.
- Secured WebSockets using a **Redis-backed ticketing system** (30s TTL) to safely bridge standard **JWT authentication**.

Vortex - Video Streaming Platform 🌀 *Spring Boot 3.5, PostgreSQL, AWS S3/DO Spaces, Spring Security, Next.js*

- Built an **async video pipeline** in **Java 21** using **ExecutorService**, eliminating timeouts via background transcoding.
- Implemented **Adaptive Bitrate streaming** using **FFmpeg filter_complex**, generating 360p/720p HLS renditions with auto quality switching.
- Parallelized S3 uploads with **CompletableFuture.allOf()**, reducing post-transcode upload time from **120s** to **20s**.
- Secured 10+ **REST API** endpoints using **JWT** and **Spring Security**, with a **GlobalExceptionHandler** to prevent data leakage.
- Hosted on **DigitalOcean** with 2GB swap space and PostgreSQL tuning, eliminating 100% of OOM crashes during transcoding.

Technical Skills

Languages: Java, C++, Python, JavaScript/TypeScript

Backend & Frameworks: Spring Boot, Microservices, Node.js, Express, Next.js, React, LangGraph

Databases & Caching: PostgreSQL, Redis, MySQL, MongoDB

Distributed Systems & Cloud: Kafka, WebSocket (STOMP), AWS S3, DigitalOcean, FFmpeg

Observability & DevOps: Prometheus, Grafana, Zipkin, Docker, Git, GitHub, Metabase

Certifications & Achievements

Data Structures, Algorithms & Algorithmic Thinking

- CodeChef:** 3★ **Coder** on CodeChef attaining highest rating of 1608 (Top 25%).
- LeetCode:** Tackled more than 600 problems, reaching a peak of 1687 (Top 15% globally).
- Codeforces:** Pupil on Codeforces. Completed over 200 challenges, peaking at 1275 (Top 60% globally).

Leadership & Extracurricular Activities

Competitive Programming (CP) Club TY Core Member

Community of Coders (COC), VJTI

Aug 2025 - Present

Mumbai, Maharashtra

Inheritance - Mentor

Community of Coders (COC), VJTI

Oct 2025 - Present

Mumbai, Maharashtra